Customer Churn Analysis and Data Preparation
Lloyds Banking Group – Data Analytics Task

1. Project Overview

Customer churn refers to customers who stop using a company's services. In the banking sector, high churn rates can lead to significant financial loss and increased costs for acquiring new customers.
The objective of this project is to analyze customer behaviour by integrating multiple datasets, performing exploratory data analysis (EDA), cleaning and transforming the data, and preparing a dataset suitable for machine learning models that predict customer churn.

2. Project Objectives

The primary objectives of this project are:

- Integrate customer data from multiple sources
- Perform exploratory data analysis (EDA)
- Identify patterns in customer behavior
- Detect potential churn indicators
- Prepare a clean dataset for machine learning
- Generate business insights for improving customer retention

3. Tools and Technologies

Tool	Purpose
Python	Data analysis and preprocessing
Pandas	Data manipulation
NumPy	Numerical computing
Matplotlib	Data visualization
Seaborn	Statistical visualization

4. Importing Libraries

The first step involves importing the necessary Python libraries for data manipulation and visualization.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

These libraries are widely used in data analysis projects to handle datasets and generate insights through visualizations.

5. Loading the Dataset

The project dataset is stored in an Excel file containing multiple sheets representing different aspects of customer data.

```
file = 'F:/Downloads/Customer_Churn_Data_Large.xlsx'

demographics = pd.read_excel(file, sheet_name='Customer_Demographics')
transactions = pd.read_excel(file, sheet_name='Transaction_History')
service = pd.read_excel(file, sheet_name='Customer_Service')
online = pd.read_excel(file, sheet_name='Online_Activity')
churn = pd.read_excel(file, sheet_name='Churn_Status')
```

Dataset Sources

Dataset	Description
Customer_Demographics	Customer personal information
Transaction_History	Records of customer purchases
Customer_Service	Customer support interactions
Online_Activity	Login and platform usage data
Churn_Status	Indicates whether the customer churned

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6. Data Integration

The datasets were merged using the **CustomerID** field to create a unified customer dataset.

```
df = demographics.merge(online, on='CustomerID')
df = df.merge(churn, on='CustomerID')
```

This step ensures that demographic information, online activity, and churn status are combined into one dataset.

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7. Transaction Data Aggregation

Customers may have multiple transaction records. To simplify the dataset, the total amount spent by each customer was calculated.

```
txn_summary = transactions.groupby('CustomerID')['AmountSpent'].sum().reset_index()
txn_summary.rename(columns={'AmountSpent':'TotalSpent'}, inplace=True)
df = df.merge(txn_summary, on='CustomerID')
```

New Feature

TotalSpent – represents the total amount spent by each customer.

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8. Customer Service Interaction Aggregation

Customer service interaction data was summarized to count how many times each customer contacted support.

```
service_count = service.groupby('CustomerID')['InteractionID'].count().reset_index()
service_count.rename(columns={'InteractionID':'ServiceInteractions'}, inplace=True)
df = df.merge(service_count, on='CustomerID')
```

New Feature

ServiceInteractions – number of customer support interactions.

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9. Exploratory Data Analysis (EDA)

Basic data structure and statistical summaries were examined.

```
df.info()
df.describe()
```

Dataset Summary

Metric	Value
Total Records	668
Total Features	11
Numeric Columns	6
Categorical Columns	4

The dataset contains demographic, behavioral, and financial information about customers.

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10. Data Visualization and Analytical Insights

Data visualization helps uncover patterns in customer demographics, financial behavior, and engagement levels.

10.1 Demographic Profile: Age Distribution Visualization

```
plt.hist(df['Age'])
plt.title('Age Distribution')
plt.tight_layout()
plt.show()
```

Purpose

This visualization identifies the **primary age segments of the customer base**.

What to Look For

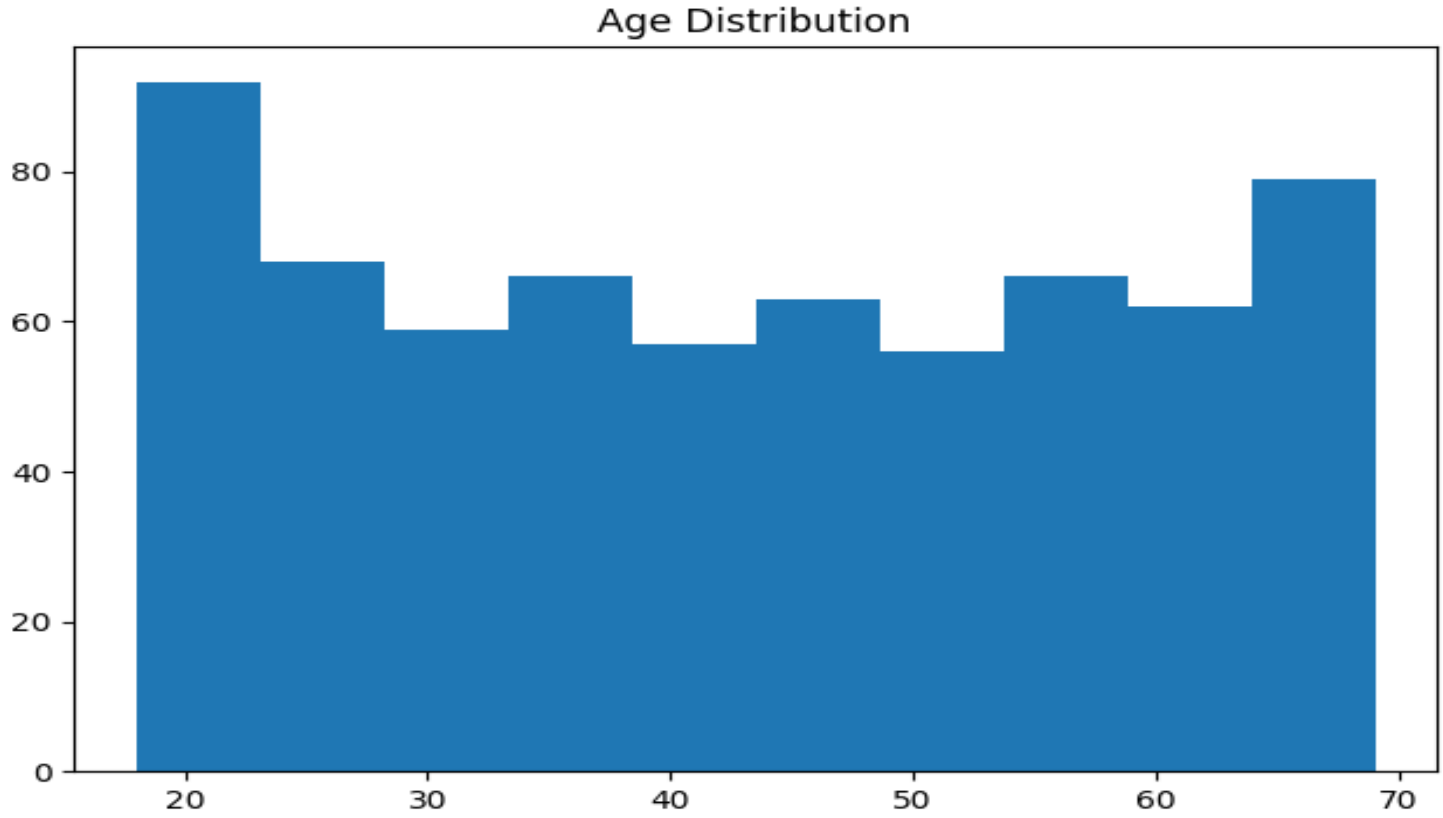
- **Normal Distribution (Bell Curve):**
  Indicates the product appeals to a wide age range.
- **Skewed Distribution:**
  Indicates the product targets a specific generation such as younger users or senior customers.

Business Insight

Understanding the peak age group helps organizations:

- Tailor marketing strategies
- Design age-appropriate financial products
- Improve digital interface accessibility

For example, if the majority of customers are aged **30–50**, banks may promote mortgage plans, investment options, and retirement planning services.



10.2 Financial Health: Transaction Spending Visualization

```
plt.boxplot(x=df['TotalSpent'])
plt.title('Transaction Spending')
plt.tight_layout()
```

```
plt.show()
```

Purpose

This visualization identifies **customer spending patterns and potential outliers**.

What to Look For

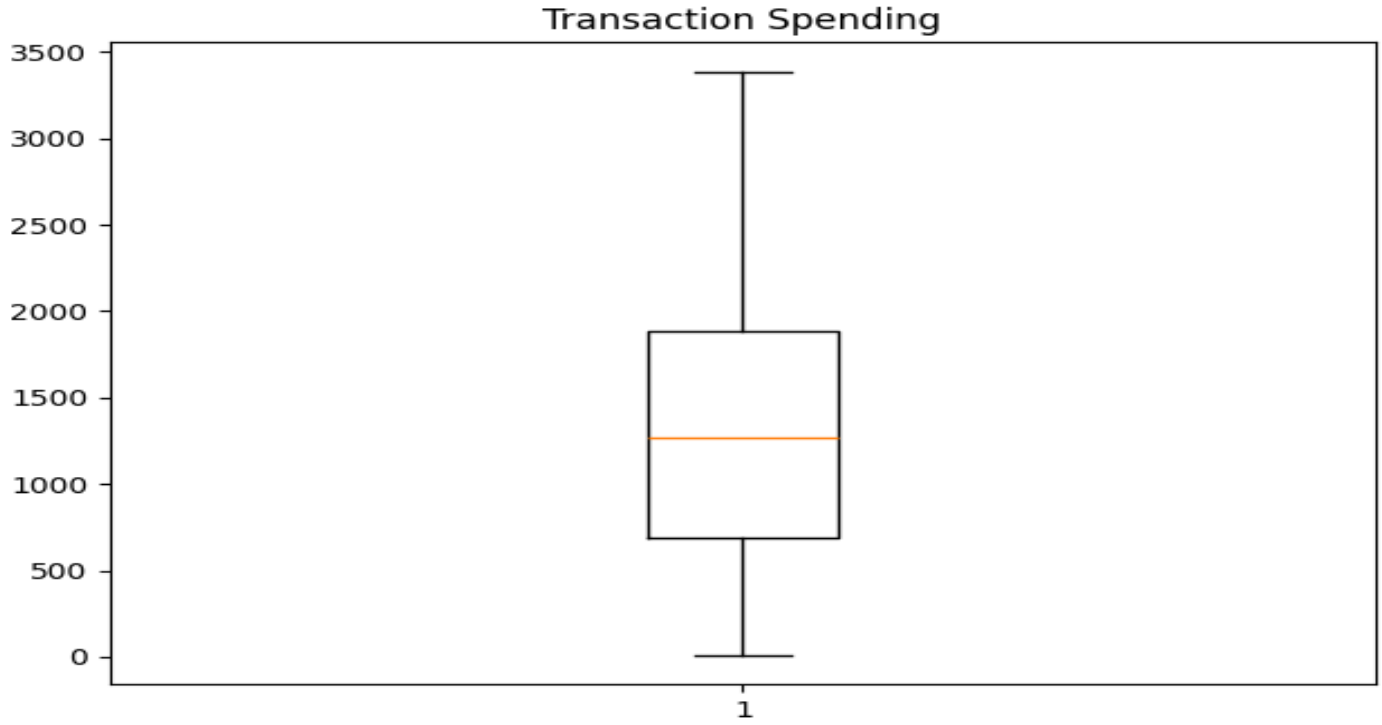
- **Median:** Typical customer spending amount.
- **Interquartile Range:** Range where most customers fall.
- **Outliers:** High-value customers who spend significantly more.

Business Insight

Outliers often represent **high-value customers (Whales)**.

If many outliers exist, the business might be **revenue-dependent on a small group of customers**.

This insight helps businesses create **VIP customer retention programs**.



10.3 Business Retention: Churn vs Non-Churn Visualization

```
sns.countplot(x='ChurnStatus', data=df)
plt.title('Churn vs Non-Churn')
plt.tight_layout()
plt.show()
```

Purpose

This visualization shows the **distribution of churned vs active customers**.

What to Look For

The ratio between:

- Customers who remain active
- Customers who have churned

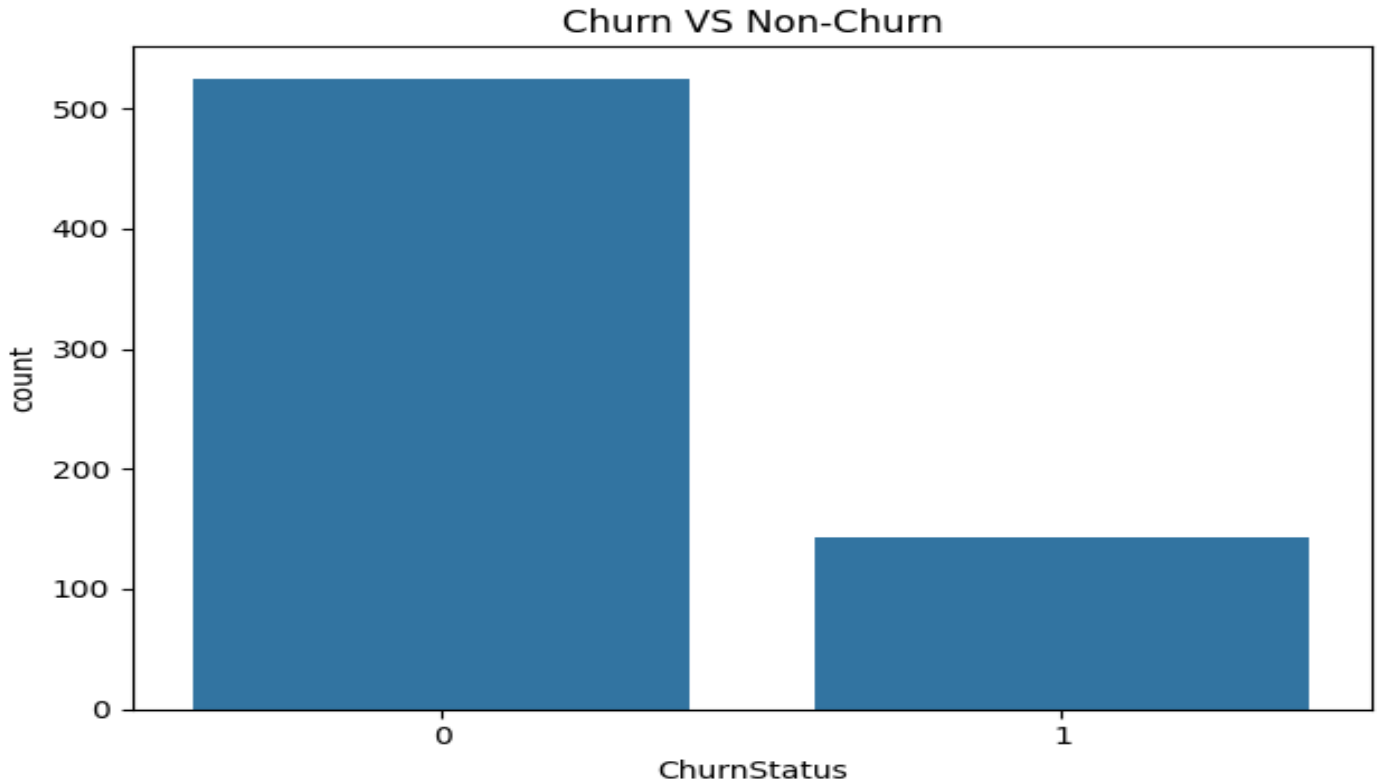
Business Insight

This is one of the **most critical indicators of business health**.

A high churn rate may indicate:

- Customer dissatisfaction
- Strong competition
- Poor service experience

Organizations must implement **customer retention strategies** such as loyalty rewards, improved support services, and personalized financial products.



10.4 Behavioral Correlation: Login Frequency vs Spending Visualization

```
plt.scatter(df['LoginFrequency'], df['TotalSpent'])
plt.xlabel('Login Frequency')
plt.ylabel('Total Spent')
plt.tight_layout()
plt.show()
```

Purpose

This visualization explores whether **customer engagement leads to higher revenue**.

What to Look For

Positive Correlation

If spending increases with login frequency, it indicates that engaged users generate more revenue.

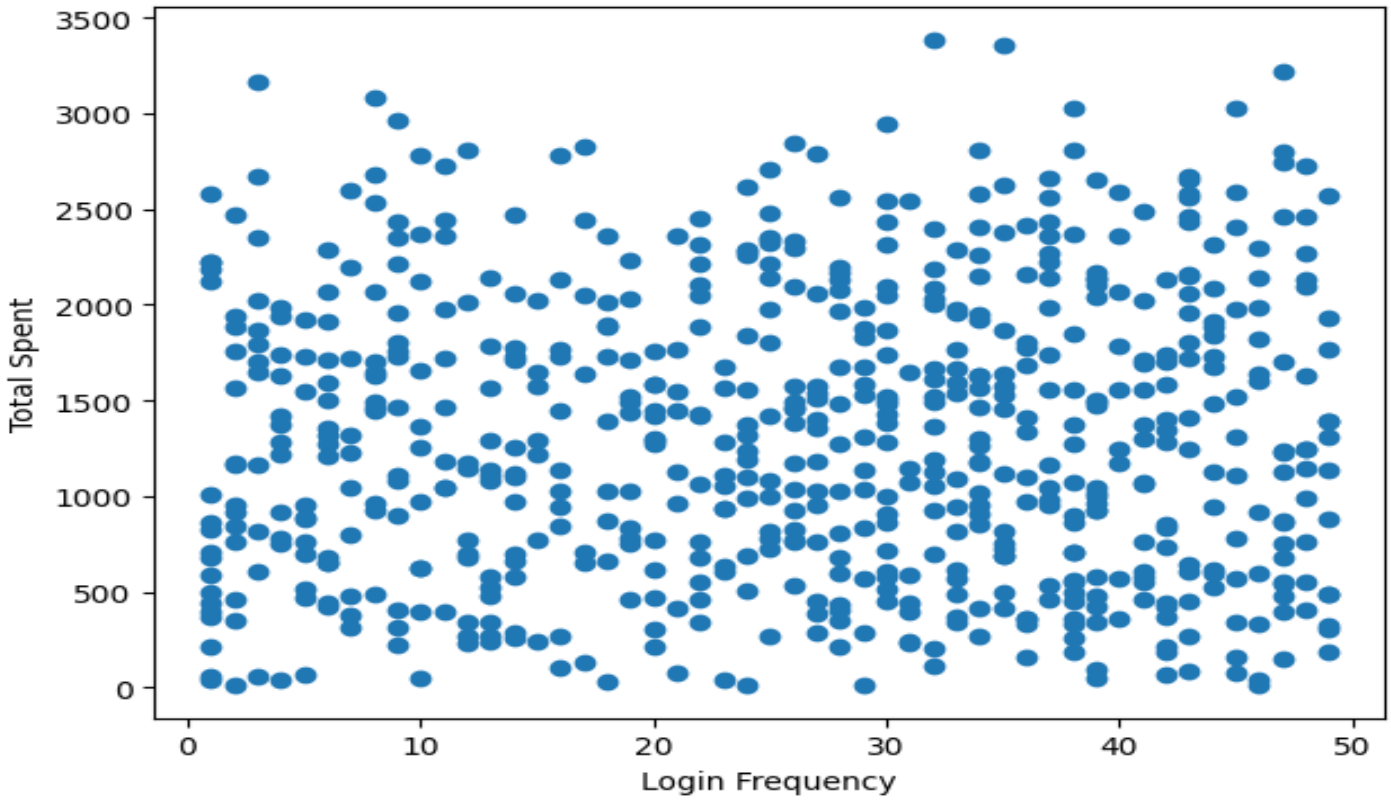
Customer Clusters

Different user segments may appear:

- High login – high spending
- High login – low spending
- Low login – high spending

Business Insight

If there is no clear relationship, it suggests that **simply increasing logins is not enough to increase revenue**. Instead, businesses must improve **session quality**, such as offering personalized recommendations and targeted financial products.



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11. Handling Missing Values

Missing values were checked using:

```
df.isnull().sum()
```

No missing values were found in the dataset.

However, precautionary handling was applied:

```
df['TotalSpent'].fillna(df['TotalSpent'].mean(), inplace=True)
df['Gender'].fillna('Unknown', inplace=True)
```

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12. Outlier Detection

Outliers were removed using the **Interquartile Range (IQR) method**.

```
Q1 = df['TotalSpent'].quantile(0.25)
Q3 = df['TotalSpent'].quantile(0.75)

IQR = Q3 - Q1

df = df[(df['TotalSpent'] >= Q1 - 1.5 * IQR) &
        (df['TotalSpent'] <= Q3 + 1.5 * IQR)]
```

Removing outliers improves model accuracy and reduces bias.

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13. Encoding Categorical Variables

Categorical features were converted into numerical format using **One-Hot Encoding**.

```
df = pd.get_dummies(df, columns=['Gender','MaritalStatus','IncomeLevel','ServiceUsage'])
```

---

14. Feature Scaling

Important numerical variables were standardized using **StandardScaler**.

```
from sklearn.preprocessing import StandardScaler
scaler = StandardScaler()
```

```
df[['Age','TotalSpent','LoginFrequency']] = scaler.fit_transform(
    df[['Age','TotalSpent','LoginFrequency']]
)
```

Feature scaling ensures consistent model performance.

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15. Final Dataset Export

The cleaned dataset was saved for machine learning modeling.

```
df.to_csv('Cleaned_churn_Data.csv', index=False)
```

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16. Business Insights

Key insights derived from the analysis:

- Customer engagement plays a significant role in revenue generation.
- High-value customers contribute disproportionately to revenue.
- Churn patterns highlight the importance of retention strategies.
- Customer service interactions may indicate dissatisfaction.

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17. Conclusion

This project successfully integrated multiple customer datasets and prepared a clean dataset suitable for churn prediction modeling.

The analysis identified key patterns in customer demographics, financial behaviour, and engagement levels.

The final dataset can be used to build machine learning models that predict customer churn, helping banks implement proactive retention strategies and improve customer satisfaction.